

Elliott Wave Momentum Strategy — Backtest Report (Corrected)

❏ **CORRECTION NOTICE (2026-07-15).** An earlier version of this report cited **CAGR 16.4% / Calmar 4.27**. An independent audit found **two look-ahead-bias bugs** in the backtest pipeline: (1) the regime/trend gates read the *still-forming* daily bar's final close (up to 24h of future data), and (2) entry fills were granted *inside* the signal bar, before the signal was knowable. Both have been fixed. The figures below are the **causal, corrected results**. They remain in-sample and should be treated as preliminary — the strategy is under review and is NOT deployed.

Instrument universe: 10 major crypto perpetual futures (Binance USDM)

Timeframe: 4-hour signals, 30-minute execution

Backtest period: 2020-01-31 → 2025-11-21 (5.81 years of trading activity)

Data coverage: through 2026-06-04 (current). No trades since Nov 2025 — the strategy is in a bear-market cash position (see §4), not a data gap.

Starting capital: \$100,000

Costs modeled: 8 bps fee + 5 bps slippage per side (round-trip ~26 bps)

1. Strategy Description

A **systematic Elliott Wave + Fibonacci** strategy for crypto futures, built around **Wave 3 — the longest and most impulsive wave** in the Elliott Wave sequence. It detects high-momentum impulse setups and uses **Fibonacci retracement levels** to time entries, positioning to capture the Wave 3 thrust. Winners are managed with a trailing exit; a predefined invalidation level caps each loss. A regime filter stands the strategy aside in confirmed bear markets, and position size adapts to recent performance.

2. Headline Performance (corrected, causal)

Metric	Value
Total return	+57.8%
CAGR	+8.17%
Maximum drawdown	−4.11%
Calmar ratio (CAGR ÷ MaxDD)	1.99
Sharpe ratio (annualized)	1.21
Sortino ratio (annualized)	10.41
Profit factor	1.93
Win rate	37.0%
Mean return per trade	+0.38 R

Metric	Value
Number of trades	416 (298 long / 118 short)

Honest read. After removing the look-ahead bias, the risk-adjusted profile is modest (Calmar ≈ 2.0 , Sharpe ≈ 1.2). The result is also **highly concentrated**: the top ~20 of 416 trades account for effectively all net profit — remove them and the remaining trades are net-negative in aggregate. Profit clusters in the 2021 and 2024 bull markets. This is the signature of a **weak, fat-tailed trend signal, not a robust all-weather edge** (see §5).

3. Year-by-Year Performance (corrected)

Year	Trades	Return	Market regime
2020 (partial)	48	+1.8%	Post-COVID recovery
2021	97	+22.4%	Bull market
2022	4	+0.9%	Bear — stood aside
2023	95	+7.0%	Recovery
2024	105	+16.2%	Bull market
2025 (partial)	67	+1.7%	Topping → bear (cash from Nov)
2026 (YTD)	0	0.0% (in cash)	Bear — standing aside (data current to Jun)

Profit is concentrated in the two bull years (2021, 2024); every other year is roughly flat.

4. Characteristic: A Bull-Market-Dependent Signal

The strategy only produces meaningful returns in trending bull markets (2021, 2024). In bears and choppy conditions it is flat to negligible, and the regime filter pulls it into cash during confirmed downtrends (2022, and currently since Nov 2025). Returns should be understood **across a full cycle**, and any single year may be flat or a "waiting" year. As of this report, the strategy is in cash, standing aside in the current bear.

5. Limitations & Caveats (read this section)

- **Corrected for look-ahead, but still in-sample.** The figures above are causal, but the strategy's thresholds, the regime gate, the coin universe, the sizing scheme, and the bear stand-down rule were all selected on this same 2020–2025 history. Residual overfitting is likely; the true out-of-sample forward edge is probably **lower than 8%, possibly negligible**.
- **Extreme concentration.** ~20 trades drive all net profit; the single best trade is +30 R. Strip a handful of lucky trend captures and the edge disappears.
- **Independent cross-check.** A separate, from-scratch causal implementation of the same 1-2-3 impulse concept found **weak-to-no statistically significant edge, and negative performance since 2023**.

- **Not deployed.** There is no validated live edge and no live capital allocated. This report documents a research result under review, not an investable track record.
- **Backtest assumptions.** Realistic fees (8 bps) + slippage (5 bps); fills validated on 30-minute data; **no look-ahead** after the 2026-07-15 fixes.

Systematic Elliott Wave momentum strategy — research backtest, corrected for look-ahead bias 2026-07-15. In-sample, preliminary, under review — NOT a validated or deployed strategy. Past performance does not guarantee future results.